ST959 FINANCIAL STATISTICS
TERM PROJECT

HEMA SRIKAR ANKEM

5716485

MSC MATHEMATICAL FINANCE

UNIVERSITY OF WARWICK

DEPARTMENT OF STATISTICS

January 2026

Contents

Contents ii

List of figures iii

List of tables iii

Declaration iv

Abstract v

List of abbreviations vi

1 STOCK PRICE ANALYSIS 1

1.1 Stock Prices and Logarithmic Returns of the Stock . . . 1

1.1.1 Stationary Time Series Data . . . 1

1.1.2 Parameter μ . . . 2

1.1.2.1 Consistency of μ: . . . 3

1.1.2.2 Asymptotic Normality of μ: . . . 3

1.1.2.3 Confidence Interval of μ: . . . 3

1.1.3 Parameter σ . . . 3

1.1.3.1 Consistency of σ: . . . 4

1.1.3.2 Confidence Interval of σ . . . 4

1.1.4 Stock Analysis . . . 4

1.1.5 Plots . . . 6

1.2 Losses on the Risk and Value at Risk . . . 7

1.2.1 Portfolio Value . . . 7

1.2.2 Value at Risk . . . 7

1.2.3 Historical Methods and Monte Carlo Simulations . . . 8

1.3 Conclusion . . . 8

2 LINEAR REGRESSION MODEL 9

2.1 Data Analysis . . . 9

2.1.1 Plots . . . 10

2.2 Simple Linear Regression Models . . . 12

2.3 Multi Variable Linear Regression Model . . . 13

2.3.1 Best Model . . . 14

2.4 Prediction using MLRM . . . 15

2.5 Conclusion . . . 15

Bibliography 16

A Theorems and Derivations 17

B Problem 1:R Script 20

List of figures

1.1 Stock Analysis Plots 6

2.1 Histograms of Explanatory Variables 10
2.2 Quantile Quantile plots of Explanatory Variables 11
2.3 Comparison of subset selection metrics for linear regression models 15

List of tables

1.1 Estimators and their statistical properties for the log-return model 4
1.2 Confidence Intervals for the Parameters of Stock A 4
1.3 Normality and Stationary Test 5
1.4 VaR results for Stock A at 95%, 99% and 99.9% confidence levels 8

2.1 Shapiro-Wilk Normality Test on Factors 9
2.2 Factor A and Factor B Regression Results 12
2.3 Factor C and Factor D Regression Results 12
2.4 Regression Results on Factor E 13
2.5 VIF values of Factors 13
2.6 BIC Score of different models 14
2.7 Multi Regression Results with all Factors 14
2.8 Multi Regression Results of Best Model 14

Declaration

This report is submitted to the University of Warwick in support for ST959: Financial Statistics module. It has been composed by myself and has not been submitted in any previous application for any degree. **No form of Artificial Intelligence** has been used in producing any parts of this report, which includes the R programming script as well. Several R packages has been used in this project which includes, Poltly [5], Stargazer [2], the R syntax and ideas are taken from examples mentioned in their documentation. All the techniques and functions used in both the report and the R script have been implemented after thoroughly studying the lecture notes, seminar and assignment resources and the R documentation available on the internet.

The back cover page was taken from the website of Department of Statistics, University of Warwick

Abstract

This first part of this report establishes an empirical analysis on an financial asset, followed by statistical analysis on a simulated data in finding a best Multi Linear regression model to explain a variable based on several explanatory variables. The Empirical analysis on the asset involves studying the historic data of stock prices using the Geometric Brownian Motion model, its logarithmic returns, the distribution of logarithmic returns, and Risk measurement. We are also interest in the estimators of the parameters of the model, their statistical properties and confidence intervals. We start our analysis with the theoretical aspects of the model and follow it with applications to our stock prices. We conclude our discussion by evaluating the Value at Risk for our asset by using the Historical methods and Monte Carlo simulations for a better inference.

The Second part of this report, deals with estimating the profits or losses of an insurance company based on five possibles factors. We start with finding the significant explanatory variables in explaining the explained variable among all the factors, by using simple linear regression of each factor against the explained variable. With this information, we compare the two multi linear regression model, the model with all factors and the model with significant factors to determine the best model. This will be followed by evaluating the Bayes Information Criterion (BIC) score for all possible models and confirming that our best multi linear regression model is has the lowest BIC score among all the models. Our study concludes with predicating the loss or profit of the insurance company based on a observation using our best model.

List of abbreviations

BIC	Bayes Information Criterion
C.L.T	Central Limit Theorem
CDF	Cumulative Distribution Function
CMT	Continuous Mapping Theorem
i.i.d	Independent and Indentically Distributed
KDE	Kernal Density Estimate
MLR	Multi Linear Regression
SLR	Simple Linear Regression
Std Dev	Standard Deviation
VaR	Value at Risk
VIF	Variance Inflation Factor

1 STOCK PRICE ANALYSIS

We start our theoretical analysis on the given Geometric Brownian Motion model by determining the distribution of the logarithmic returns and properties of the time series data. We propose a suitable estimator for the parameters of the model, study their statistical properties and find their confidence intervals. With the results from the theoretical analysis we can conducted empirical analysis on the historical data of stock prices. We perform various tests to support our assumptions, both quantitatively and visually, and then discuss about the notion of portfolio value, risk measure and simulate Value at Risk using Historical and Monte Carlo methods at different confidence intervals.

1.1 Stock Prices and Logarithmic Returns of the Stock

The model for logarithmic returns of an asset is given by,

$$\begin{aligned} X_k &:= \ln\left(\frac{S_t}{S_{t-\Delta}}\right) = \left(\mu - \frac{\sigma^2}{2}\right)\Delta + \sigma Y_t \\ X_{k-1} &:= \ln\left(\frac{S_{t-\Delta}}{S_{t-2\Delta}}\right) = \left(\mu - \frac{\sigma^2}{2}\right)\Delta + \sigma Y_{t-\Delta} \\ &\vdots \\ X_1 &:= \ln\left(\frac{S_{t-(k-1)\Delta}}{S_{t-k\Delta}}\right) = \left(\mu - \frac{\sigma^2}{2}\right)\Delta + \sigma Y_{t-(k-1)\Delta} \end{aligned} \tag{1.1}$$

- S_t = Price of a given share at time t
- Δ = Fixed time interval (one day or one week or one month)
- $Y_t, Y_{t-2\Delta}, \ldots, Y_{t-(k-1)\Delta}$ are assumed to be Independent and Indentically Distributed (i.i.d) random variables following a normal distribution with mean (μ) 0 and variance (σ^2) Δ
- $X_1, X_2, \ldots, X_k$ = log-returns on the stock

As we follow frequentist approach for this model, we consider the parameters μ and σ to be deterministic parameter in our analysis and $\mu \in \mathbb{R},\ \sigma > 0$

1.1.1 Stationary Time Series Data

A (square-integrable) stochastic process (X_i), with $i \in \{1, 2, \ldots, k\}$ is called **Stationary** if the following conditions are satisfied:

1. The expectation $\mathbb{E}(X_i)$ does not depend on time
2. The variance $\mathrm{Var}(X_i)$ does not depend on time
3. For each i, j with $i \neq j$, the covariance $\mathrm{Cov}(X_i, X_j)$ does not depend on time

Lets take the equation,

$$X_k := \ln\left(\frac{S_t}{S_{t-\Delta}}\right) = \left(\mu - \frac{\sigma^2}{2}\right)\Delta + \sigma Y_t$$

We know that Y_t follows a normal distribution with mean 0 and variance Δ, $\Rightarrow\ Y_t \sim \mathcal{N}(0, \Delta)$. By we rearranging the above equation, we get

$$Y_t = \frac{X_k - \left(\mu - \frac{\sigma^2}{2}\right)\Delta}{\sigma} \sim \mathcal{N}(0, \Delta) \tag{1.2}$$

We know that if a random variable X follows $\mathcal{N}(m, \sigma^2)$. Then, the random variable Z defined by $Z := \frac{X-m}{\sqrt{\sigma^2}}$ follows $\mathcal{N}(0, 1)$. By transforming the above equation (1.2) we get,

$$\frac{X_k - \left(\mu - \frac{\sigma^2}{2}\right)\Delta}{\sigma\sqrt{\Delta}} \sim \mathcal{N}(0, 1) \tag{1.3}$$

We can also write the distribution of X_t from the above equation, which is given by,

$$\boxed{X_k \sim \mathcal{N}\left(\left[\mu - \frac{\sigma^2}{2}\right]\Delta, \sigma^2\Delta\right)} \tag{1.4}$$

Note

Since, $Y_t, Y_{t-2\Delta}, \ldots, Y_{t-(k-1)\Delta}$ are i.i.d random variables, by Continuous Mapping Theorem (CMT), (X_i) are i.i.d random variables as well and are normally distributed, given by equation (1.4) $\forall\ i \in \{1, 2, 3, \ldots, k-1, k\}$. The stock prices are log-normally distributed.

Therefore, the mean and variance are given by,

$$\mathbb{E}[X_i] = \left(\mu - \frac{\sigma^2}{2}\right)\Delta, \qquad \mathrm{Var}(X_i) = \sigma^2\Delta \qquad \forall\ i \in \{1, 2, 3, \cdots, k-1, k\} \tag{1.5}$$

As we can see from the above equations both Mean and Variance of (X_i) does not depend on the time. The covariance of $\mathrm{Cov}(X_i, X_j)$ is given by,

$$\mathrm{Cov}(X_i, X_j) = \mathbb{E}(X_iX_j) - \mathbb{E}(X_i)\mathbb{E}(X_j) = \mathbb{E}(X_i)\mathbb{E}(X_j) - \mathbb{E}(X_i)\mathbb{E}(X_j) = 0$$

Since X_i are i.i.d random variables. Hence the the time series of the log-returns in equation (1.1) is **Stationary Time series**.

1.1.2 Parameter μ

From the equation (A.3), we can write,

$$\mathbb{E}[\bar{X_k}] = \left(\mu - \frac{\sigma^2}{2}\right)\Delta \quad \Rightarrow \quad \mu = \frac{\mathbb{E}[\bar{X_k}]}{\Delta} + \frac{\sigma^2}{2}$$

The μ for the sample space can be written using Method of Moments, and by replacing the parameter with maximum likelihood estimators, we get,

$$\widehat{\mu} = \frac{\bar{X}_k}{\Delta} + \frac{\widehat{\sigma}^2}{2} \tag{1.6}$$

The expectation of the parameter $\bar{\mu}$ is given by,

$$\mathbb{E}(\bar{\mu}) = \frac{\mathbb{E}(\bar{X}_k)}{\Delta} + \frac{\mathbb{E}(\bar{\sigma}^2)}{2} \quad = \mu - \frac{\sigma^2}{2} + \frac{\mathbb{E}(\bar{\sigma}^2)}{2} \quad = \mu - \frac{1}{2}\left[\mathbb{E}(\bar{\sigma}^2) - \sigma^2\right]$$

Therefore we get the expectation of sample parameter $\bar{\mu}$ as,

$$\boxed{\mathbb{E}(\widehat{\mu}) = \mu + \frac{1}{2}\left[\mathbb{E}(\widehat{\sigma}^2) - \sigma^2\right]} \tag{1.7}$$

We can see that from the equation (1.7), expectation of sample parameter $\widehat{\mu}$ depends on the sample size as the $\mathbb{E}(\widehat{\sigma}^2)$ is not always equals to σ^2. Only when the sample size is close to true population size, i.e when $k \to +\infty$.

- $\mathbb{E}(\widehat{\mu}) = \mu$, only when the value of $k \to +\infty$ or in reality when the value of k is very large.
- For finite k values, the estimator $\widehat{\mu}$ is biased.

1.1.2.1 Consistency of μ:

$$\bar{X}_k = \frac{1}{k}\sum_{i=1}^{k} X_i \xrightarrow[k\to+\infty]{\text{probability}} \left(\mu - \frac{\sigma^2}{2}\right)\Delta \qquad \text{: Weak Law of Large Numbers}$$

$$\widehat{\sigma}^2 \xrightarrow[k\to+\infty]{\text{probability}} \sigma^2$$

From the above two equations, and using CMT we get,

$$\widehat{\mu} \xrightarrow[k\to+\infty]{\text{probability}} \mu \tag{1.8}$$

So, from equation (1.8) we can conclude that estimator $\widehat{\mu}$ is **consistent estimator** for μ

1.1.2.2 Asymptotic Normality of μ:

Using Central Limit Theorem (C.L.T), on distribution of X_i equation (1.4) we can write,

$$\sqrt{k}\left(\bar{X} - \left(\mu - \frac{\sigma^2}{2}\right)\Delta\right) \xrightarrow{\text{Distribution}} \mathcal{N}(0, \sigma^2\Delta)$$

Using equation (1.6), results in $\bar{X}_k = \left(\widehat{\mu} - \frac{\widehat{\sigma}^2}{2}\right)\Delta$ and substituting it in the above result, we get

$$\sqrt{k}(\widehat{\mu} - \mu) \xrightarrow{\text{Distribution}} \mathcal{N}\left(0, \frac{\sigma^2}{\Delta}\right) \tag{1.9}$$

Therefore, we can conclude that estimator $\widehat{\mu}$ is **Asymptotically normal** for parameter μ.

1.1.2.3 Confidence Interval of μ:

Using the Asymptotic property section 1.1.2.2 and Slutsky's Theorem (chapter A), we can get the confidence interval for parameter μ. From the equation (A.1), we get

$$Z_n := \frac{\sqrt{k\Delta}(\widehat{\mu} - \mu)}{\widehat{\sigma}} \sim \mathcal{N}(0, 1)$$

Using the equation (A.7) and the Z_n as defined above, we can write as

$$\mathbf{P}\left[-q_z\left(1 - \frac{\alpha}{2}\right) \leqslant \frac{\sqrt{k\Delta}(\widehat{\mu} - \mu)}{\widehat{\sigma}} \leqslant q_z\left(1 - \frac{\alpha}{2}\right)\right] = 1 - \alpha$$

$$\mathbf{P}\left[\widehat{\mu} - \frac{\widehat{\sigma}}{\sqrt{\Delta k}} q_z\left(1 - \frac{\alpha}{2}\right) \leqslant \mu \leqslant \widehat{\mu} + \frac{\widehat{\sigma}}{\sqrt{\Delta k}} q_z\left(1 - \frac{\alpha}{2}\right)\right] = 1 - \alpha$$

The Confidence Interval of μ is given by,

$$\left[\widehat{\mu} - \frac{\widehat{\sigma}}{\sqrt{\Delta k}} q_z\left(1 - \frac{\alpha}{2}\right),\ \widehat{\mu} + \frac{\widehat{\sigma}}{\sqrt{\Delta k}} q_z\left(1 - \frac{\alpha}{2}\right)\right] \tag{1.10}$$

1.1.3 Parameter σ

The Adjusted Sample Variance from the equation (A.6) is given by,

$$S^2 = \frac{1}{k-1}\sum_{i=1}^{k}(X_i - \bar{X})^2 \qquad \text{where} \qquad \bar{X} = \frac{1}{k}\sum_{i=1}^{k} X_i$$

Let's define $\widehat{\sigma}^2 = \frac{S^2}{\Delta}$. The expectation of $\widehat{\sigma}$ is given by,

$$\mathbb{E}[\widehat{\sigma}^2] = \frac{1}{\Delta}\mathbb{E}[S^2] = \sigma^2.$$

Therefore we can conclude that, $\widehat{\sigma}^2 = \frac{S^2}{\Delta}$ is **unbiased estimator** for σ^2

1.1.3.1 Consistency of σ:

From equation (A.2), and substituting Variance = $\sigma^2\Delta$ we get,

$$\frac{(k-1)S^2}{\sigma^2\Delta} \sim \chi^2_k \qquad \Rightarrow \qquad \boxed{\frac{(k-1)\widehat{\sigma}^2}{\sigma^2} \sim \chi^2_k} \tag{1.11}$$

$$S^2 \xrightarrow[k\to+\infty]{\text{probability}} \sigma^2\Delta \qquad : \text{Weak Law of Large Numbers and CMT}$$

$$\text{Hence,} \qquad \widehat{\sigma}^2 = \frac{S^2}{\Delta} \xrightarrow{\mathbb{P}} \sigma^2 \tag{1.12}$$

Therefore, we can conclude that $\widehat{\sigma}$ is a **consistent estimator** for σ^2

1.1.3.2 Confidence Interval of σ

We know the distribution $\widehat{\sigma}^2$ from the equation (1.11) and using the result from the equation (A.7) but when the Z_n χ^2 distribution, we can write,

$$\mathbf{P}\left[\chi^2_{\alpha/2,k} \leqslant \frac{(k-1)\widehat{\sigma}^2}{\sigma^2} \leqslant \chi^2_{1-\alpha/2,k}\right] = 1-\alpha. \qquad \Rightarrow \qquad \mathbf{P}\left[\frac{(k-1)\widehat{\sigma}^2}{\chi^2_{1-\alpha/2,k}} \leqslant \sigma^2 \leqslant \frac{(k-1)\widehat{\sigma}^2}{\chi^2_{\alpha/2,k}}\right] = 1-\alpha$$

Since we have considered that $\sigma > 0$, the confidence interval for σ is given by,

$$\left[\sqrt{\frac{(k-1)\widehat{\sigma}^2}{\chi^2_{1-\alpha/2,k}}}, \sqrt{\frac{(k-1)\widehat{\sigma}^2}{\chi^2_{\alpha/2,k}}}\right] \tag{1.13}$$

Summary: The Statistical properties of the estimators choosen for the parameters are given below,

Parameter	Estimator	Properties
μ	$\widehat{\mu} = \frac{\bar{X}}{\Delta} + \frac{\widehat{\sigma}^2}{2}$	• Unbiased estimator for large k but biased for small k, i.e the estimator is Asymptotically unbiased. • Consistent estimator • Asymptotically normal
σ	$\widehat{\sigma}^2 = \frac{S^2}{\Delta}$	• Unbiased estimator for all k • Consistent estimator • Asymptotically normal, as χ^2 distributions approach normal distribution for large degrees of freedom

Table 1.1: *Estimators and their statistical properties for the log-return model*

1.1.4 Stock Analysis

The realization of the parameters of the given model and their confidence intervals are give by,

Parameter	Parameter Realisation	Confidence Interval	Numerical Result($\alpha = 5\%$)
μ	$\widehat{\mu} = 0.3225$	$\left[\widehat{\mu} - \frac{\widehat{\sigma}}{\sqrt{\Delta k}} q_z\left(1-\frac{\alpha}{2}\right), \widehat{\mu} + \frac{\widehat{\sigma}}{\sqrt{\Delta k}} q_z\left(1-\frac{\alpha}{2}\right)\right]$	$[-0.3349,\ 0.9710]$
σ	$\widehat{\sigma} = 0.4716$	$\left[\frac{(k-1)\widehat{\sigma}^2}{\chi^2_{1-\alpha/2,\,k}}, \frac{(k-1)\widehat{\sigma}^2}{\chi^2_{\alpha/2,\,k}}\right]$	$[0.4436,\ 0.5023]$

Table 1.2: *Confidence Intervals for the Parameters of Stock A*

Since we have considered one trading day, the value of Δ will be taken as **1/256**, which is the fraction of number of trading days available in a year. The following test results allows us to validate our theoretical assumptions.

Test	Results	Inference
Shapiro-Wilk Normality		The null-hypothesis(H_0) of Shapiro-Wilk Normality test is that, the data is Normally distributed.
	W = 0.809	We know that when W = 1, the data is perfectly normal. From the value of W, we can see it is close to 1, so there is only slight deviation from normality
	p-value $< 2.2 \times 10^{-16}$	But with the p-value, we have to reject the null-hypothesis. So, the log returns are (X_i) **not normally distributed**. This can be also verified visually from the figure 2.1a, which has clearly has Kernal Density Estimate (KDE) deviated from normal fit.
Kolmogorov-Smirnov	D = 0.11494	The normal distribution is the reference distribution used in this test. The D value is the maximum difference between reference CDF and the CDF of our data
	p-value = 3.86×10^{-6}	The null-hypothesis(H_0) is that, the data is comes from the reference distribution. From the p-value, we can under that, it is not significant. So, we reject the null-hypothesis(H_0)
	alternative hypothesis: two-sided	Thus, we can confirm that that log returns are **not normally distributed**.
Box-Pierce	χ-squared = 2.4749	The null hypothesis,H_0: The time series data behave like a white noise upto several lags and its auto correlations is zero
	df = 5	
	p-value = 0.7803	From the p-value, we can understand that H_0 is acceptable. So, the time series data is **uncorrelated**.
Ljung-Box	χ-squared = 2.499	The null hypothesis,H_0: The time series data behave like a white noise upto several lags and its auto correlations is zero This test also deals with the issue with fitness of the sample.
	df = 5	
	p-value = 0.7766	From the p-value, we can understand that H_0 is acceptable. So, the time series data is **uncorrelated**.

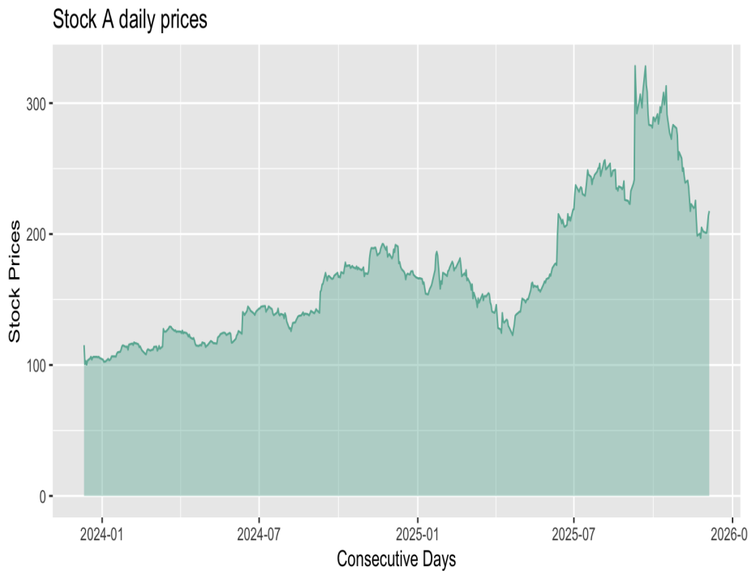
Table 1.3: *Normality and Stationary Test*

Stationarity of the Time Series Data: From the table 1.3, we can conclude that, the Stock Prices A given in our data set is **not normally distributed** and also are **not correlated**. This can be also visually obsered from the the figure 1.1d of log returns, which has alot of deviations from the quantile reference line. Using the R program, we get the value of $\mathbb{E}[X_i] = 0.00128$ and $\text{Var}(X_i) = 0.00088$. The Covariance matrix (of size 498×498) is a **diagonal matrix** with elements equal to the $\text{Var}(X_i)$. Hence the the time series of the log-returns in equation (1.1) based on the our data is **Stationary Time- series**.

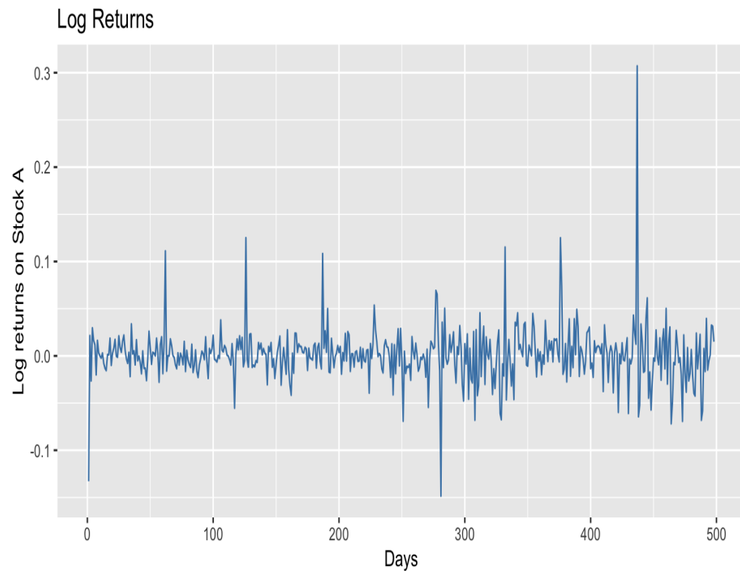
Independent and Identically Distributed Analysis: From the figure 1.1f we can see almost a circular cloud like scatter plot, as explained by Meucci [4] that a circular scatter plot explains that market invariants are i.i.d and circularity indicates true market invariance. The Identical distribution nature can also be verified visually from the figure 1.1e, as both histograms almost look identical. The first half represents the first half of the whole data and second half for the remaining data. Even though there are few deviations in the plots, we neglect the deviations and consider them as outliners and conclude our analysis based on the majority of the pattern or trend.

Therefore, we can conclude that, the time series data is not normally distributed, but i.i.d random variables with no correlation.

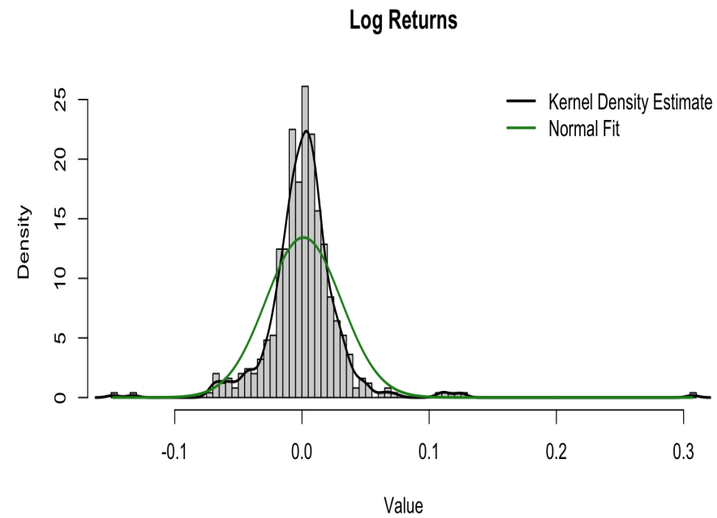
1.1.5 Plots



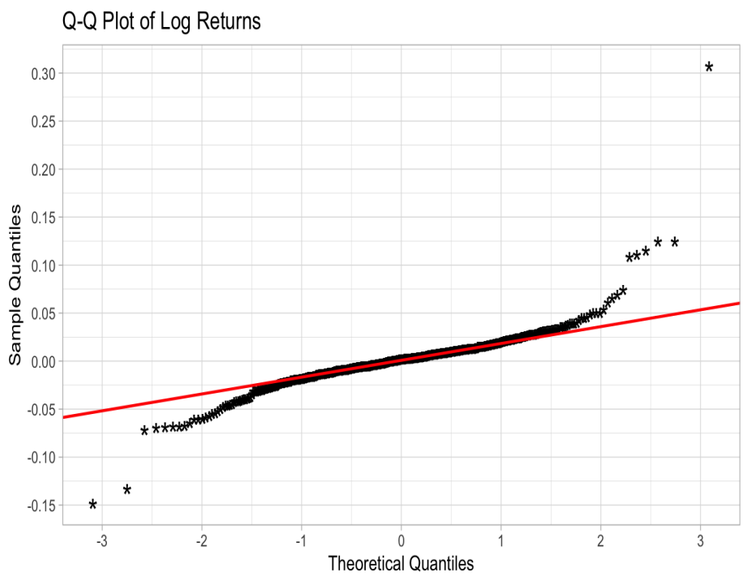
(a) *Price History of the Asset*



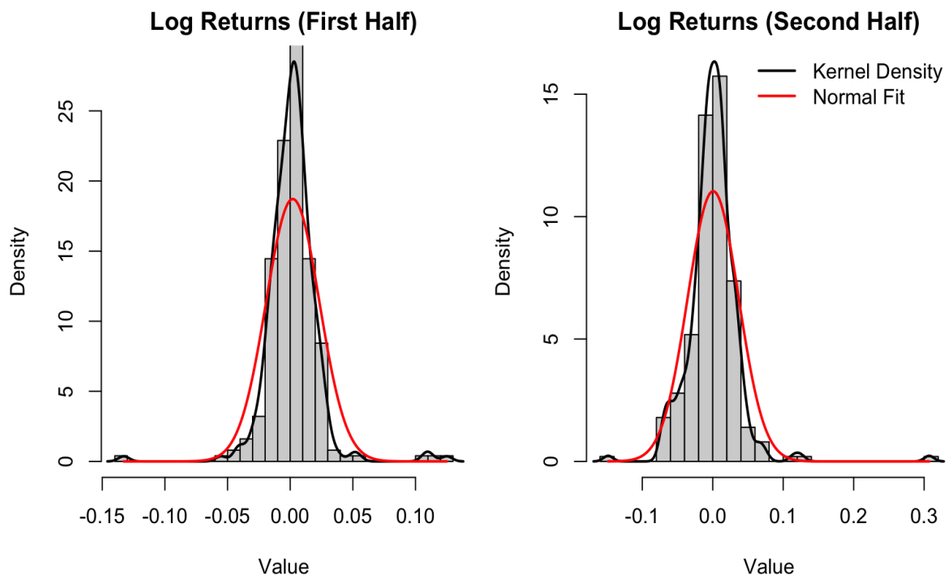
(b) *Log Returns*



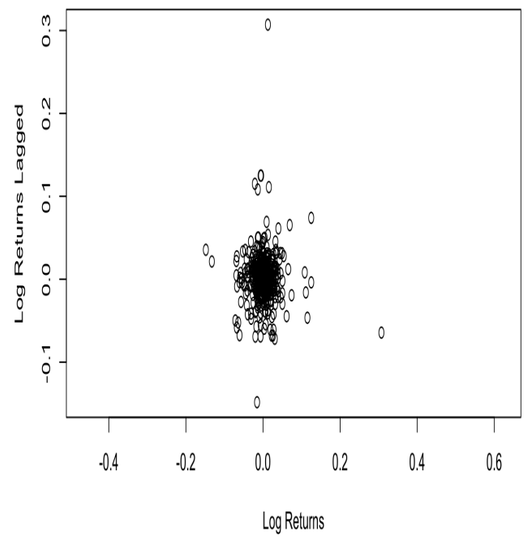
(c) *Histogram plot of Log Returns*



(d) *Quantile-Quantile plot of Log Returns*



(e) *Histogram of first and second half of Log Returns*



(f) *Meucci Visual Analysis*

Figure 1.1: *Stock Analysis Plots*

> ***Note***
>
> The above plots are only for visually analysis. We certainly reply upon the test results for the final conclusions. But for the visual analysis, we can neglect the small deviations and outliners to fit our theoretical analysis to our data and the model in a whole.

1.2 Losses on the Risk and Value at Risk

1.2.1 Portfolio Value

The portfolio value at any given time is the total value of our assets, calculated by summation of all our investments value, which is multiplying the number of positions we hold of that asset by the current price of the asset.

- ζ_t^d = Trading Strategy for asset d at time t and S_t^d = Stock price of asset d at time t
- V_t = Wealth or the portfolio value at the time t
- G_t = Gains/Losses till the time t. If the G_t found to be negative then our portfolio incurred losses and if it is positive then profits.

The decision of taking long position or taking short position and the number of positions on an asset will be contained in trading strategy ζ_t^d. We consider the V_t to be random variable [1], which is given by $V_t = \zeta_t \cdot S_t$. The profit over time time period from t to $t + \Delta$ is given by

$$V_{t+\Delta} - V_t = \sum_{d=1}^{n} \left(\zeta_{t+\Delta}^d \cdot S_{t+\Delta}^d - \zeta_t^d \cdot S_t^d\right)$$

At the end of the tth trading period, the Gain process is give by,

$$G_t = \sum_{i=1}^{k} \sum_{d=1}^{n} \zeta_i^d \cdot \left(S_{i+1}^d - S_i^d\right) \tag{1.14}$$

For simplicity let us consider only one asset, as given in our dataset, so we can drop the d variable from our analysis. Let, $L_{t+\Delta}$ denote the loss over the time period from t to $t + \Delta$, given by $L_{t+\Delta} := -(V_{t+\Delta} - V_t)$ then the probability distribution of $L_{t+\Delta}$ is called **Loss distribution**. Negative sign in $L_{t+\Delta}$ make losses to be positive in our risk theory.

1.2.2 Value at Risk

Risk Measure: A risk measure is a mapping from Loss L to a real number. The number indicates the riskiness and the mapping should be monotonically increasing i.e if $L_{t_i} < L_{t_{i+1}}$ then the mapping ρ, should follow $\rho(L_{t_i}) < \rho(L_{t_{i+1}})$. The risk measure considers risk factors while mapping the loss and the choice of risk factors is a modeling issue and depends on the portfolio. The loss distribution is a function a portfolio value which depends on the risk factors. Since, risk factors changes over the time, we can only estimate the loss distribution based on a explicit parameter, which measures the risk factor changes over the time.

Value at Risk: The maximum losses incurred on a portfolio can be our whole capital or infinite theoretically if we considered short positioning. So, the maximum loss or the maximum risk holding a portfolio notion does not make sense mathematically. To replace this maximum loss, we introduce Value at Risk (VaR) defined as "maximum loss that is not exceeded with a given high probability" by Mc.Neil [3] The VaR of a portfolio at a given confidence level α, with loss L is the smallest number l, that the probability of loss L exceeds l is no larger than $1 - \alpha$ and it is denoted by VaR_α

$$\mathrm{VaR}_\alpha = \mathrm{VaR}_\alpha(L) = \inf\{l \in \mathbb{R} : \mathbf{P}(L > l) \leqslant 1 - \alpha\} \tag{1.15}$$

If loss distribution follows a normal distribution $\mathcal{N}(m, \sigma^2)$, then VaR is given by,

$$\mathrm{VaR}_\alpha(L) = m + \sigma\Phi^{-1}(\alpha) \tag{1.16}$$

where Φ denotes the cumulative distribution function of standard normal distribution $\mathcal{N}(0, 1)$

1.2.3 Historical Methods and Monte Carlo Simulations

Historical Simulation: Since the loss distribution depends on an explicit parameter, the historical simulation method takes the historical data of the asset to find the historical changes in the risk factors to determine its empirical distribution and using this empirical distribution of risk factors, we determine the loss distribution.

Through historical simulation, we can understand what happens to our portfolio value if the historical changes of risk factors happen again in the future. As stated by Mc.Neil [3], the empirical distribution of historically simulated losses will be a consistent estimator of the loss distribution and VaR can be calculated using the from emperical distribution of historically simulated losses, which is given by the equation (1.16)

Monte Carlo simulation: As described by Mc.Neil [3], an approach to measure risk that involves the simulation of an explicit parametric model for risk factor changes is know as Monte Carlo Simulations. For large portfolios the computational cost for the Monte Carlo simulations will be enormous. Monte Carlo simulations is used to generate the realization of the explicit parameter which arose due to the changes in risk factors.

Please refer to equation (A.8), for the derivation of the loss operator that was used in the estimation of the VaR_α. Using our historical data and following the procedure given in the lecture slides of Prof. Grigorova [1], we get the following VaR results,

Method	**VaR Value** ($\alpha = 95\%$)	**VaR (%)** ($\alpha = 95\%$)	**VaR Value** ($\alpha = 99\%$)	**VaR (%)** ($\alpha = 99\%$)	**VaR Value** ($\alpha = 99.9\%$)	**VaR (%)** ($\alpha = 99.9\%$)
Historical Method	8.85	4.07%	14.32	6.57%	27.063	12.44 %
Monte Carlo Simulation	10.14	4.66%	14.31	6.58%	18.88	8.68%

Table 1.4: *VaR results for Stock A at 95%, 99% and 99.9% confidence levels*

From the table 1.4 we can see that, the historical simulation results in a higher VaR value than Monte Carlo simulation as historical method is more sensitive to past extreme losses. As we can observe from thefigure 1.1b, there are several extreme positive and negative values, these extreme cases increased the sensitivity of the historical simulation where as the Monte Carlo Simulations smoothen the extreme cases giving a lower estimate of rare extreme cases. Since our data set of stock prices has limited data points, historical simulation methodology fails to produce any reliable estimate of VaR_α.

> ***Note***
>
> The α value used in calculating the confidence intervals of parameter is the significance level, $\alpha \in (0, 1)$ and the value $(1 - \alpha)$ gives you the confidence level. Where as the α value used in the calculations of VaR is the confidence level. So VaR at $\alpha = 0.95$, gives us the value that is at risk with 95% confidence level.

In the table 1.4, the VaR value is the monetory value at risk and the VaR (%) is the percentage of our portfolio that is at risk with that given confidence level. Through our analysis, we found that, the maximum loss(VaR) our portfolio can incur with 99.9% is given by 27.063 with the historical simulation and 18.88 with the Monte Carlo Simulation. We have almost similar results with both the methodologies when the confidence level is 99%.

1.3 Conclusion

From our empirical analysis, we have found the logarithmic returns of Stock price of A does not follow a normal distribution, which is violates our assumption. It has a heavy tailed distribution, which is common with any asset prices. But, we also found that the logarithmic returns follows our assumptions that they are i.i.d random variables and uncorrelated. Through our simulations we have found that historical methodology produced higher VaR than the Monte Carlo methodology.

2 LINEAR REGRESSION MODEL

In this exercise, we study about the linear regression model of loss (Yvariable) of an insurance company with possible influence of five factors identified as Factor A, Factor B, Factor C, Factor D and Factor E. The data has over 2000 observations, with response variable taking both negative values corresponding to gains and positive values corresponding to losses.

We begin our analysis by understanding the probability distribution of all the explanatory variables. Then by performing Simple Linear Regression (SLR) on Yvariable against factor to understand the importance of each explanatory variable in explaining the Yvariable outcome. This gives us preliminary understanding of the Multi Linear Regression (MLR) model. Then we follow our study on each model and evaluate them on the Bayes Information Criterion (BIC) and Adjusted R^2-squared value to find the best model. Finally, we predict the Yvariable for a given observation of factors.

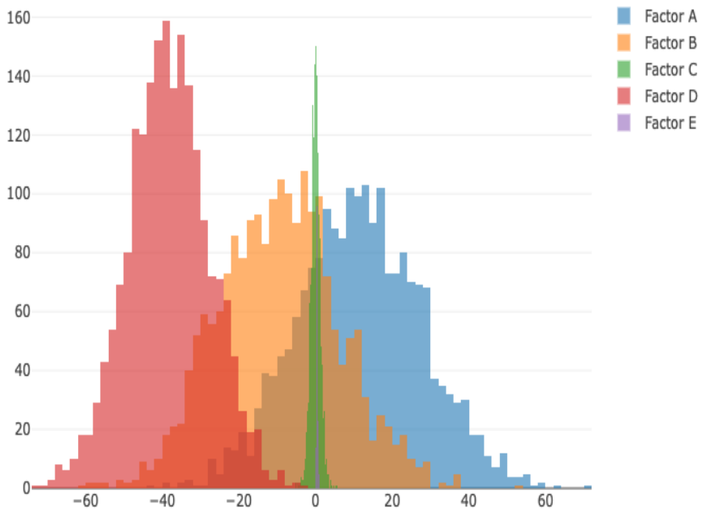
2.1 Data Analysis

Using the Shapiro-Wilk Normality test, we can understand which of the factors are normally distributed. We can also understand the normality of the factors data visually as we have done in figure 2.1. Using the Kernal Density Estimate (KDE) of factors data along with normal fit, we can observe the deviations.

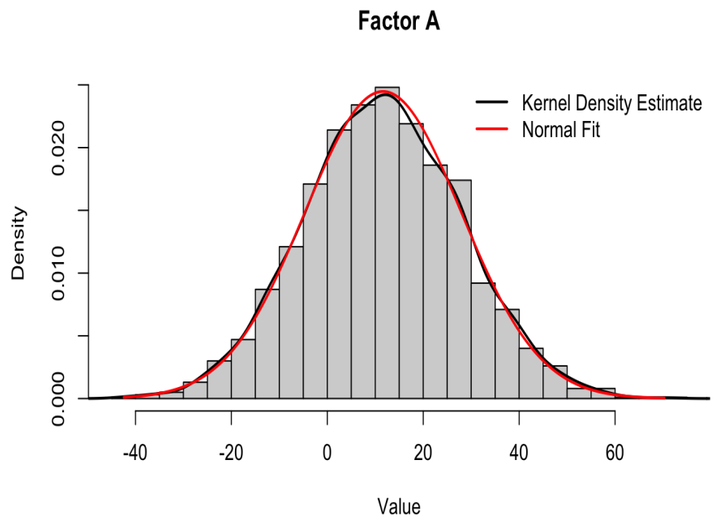
Factor	Test Results	Inference	Result
Factor A	W = 0.9997	The value of W is close to 1 indicating the data is very so slightly deviated from perfect normal distribution. We can also observe visually from the figure 2.1b that KDE has small deviations from the normal fit.	Normally Distributed
	p-value = 0.9953	The p-value is significant, which is 0.99, indicating within ~ 10% confidence level, we can accept the H_o	
Factor B	W = 0.9989	Though the W value is close to 1, we can observe visually from figure 2.1c that there are many deviations from the normal fit. But p-value helps us determining the outcome	Normally Distributed
	p-value = 0.2435	From p-value, we can understand that only with ~ 20% confidence levels, we can accept the H_0, which also supports deviations from our visual observation	
Factor C	W = 0.9918	The value of W is no longer a significant parameter for our inference. As, we can see fro figure 2.1d the KDE is steeper than normal fit and with alot of deviations.	Not Normally Distributed
	p-value = 3.343e-09	The p-value, is very small, to consider it to be significant. So, we reject the H_0	
Factor D	W = 0.9994	From the figure 2.1e, we can clearly observe that KDE has very few deviations from normal fit.	Normally Distributed
	p-value = 0.8249	With p-value being 0.82, we can conclude that, within ~ 10% confidence interval, we can accept the H_0	
Factor E	W = 0.95152	From figure 2.1f, we can clearly see that the KDE is left skewed, clearly indicating that it is not a normal distribution.	Not Normally Distributed
	p-value = 2.2e-16	Also, p-value being very small, which is insignificant, we can reject the H_0	

Table 2.1: *Shapiro-Wilk Normality Test on Factors*

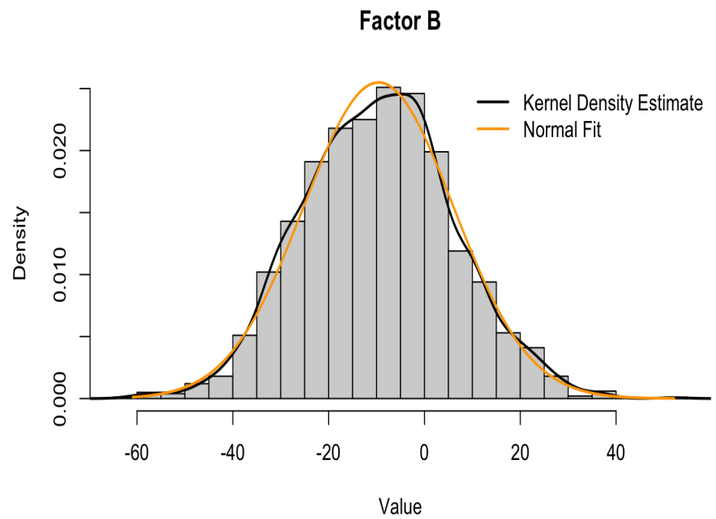
2.1.1 Plots



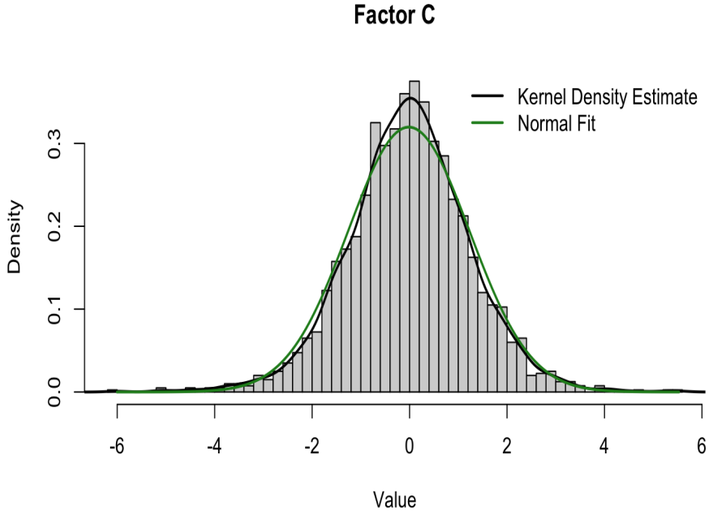
(a) *Histogram of Factors*



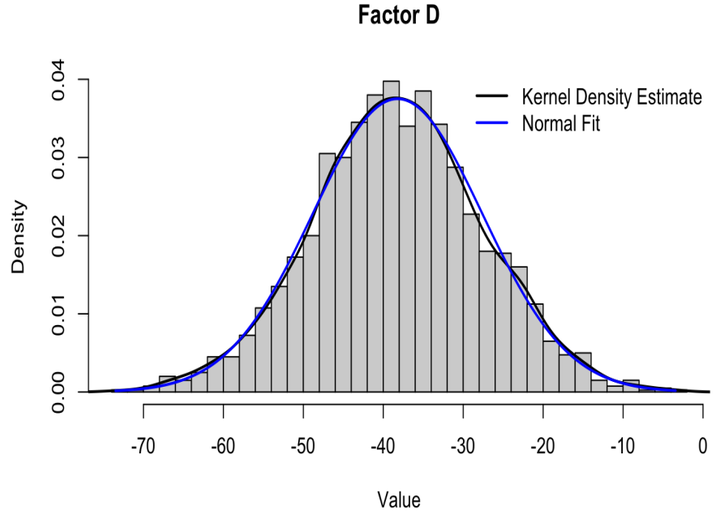
(b) *Histogram of Factor A*



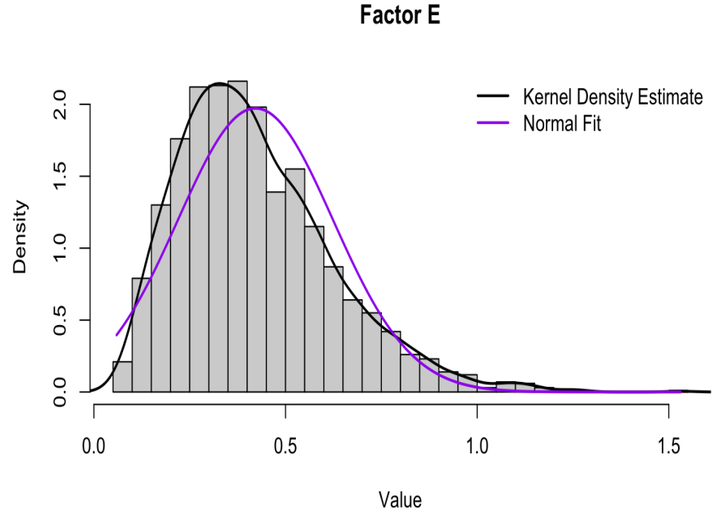
(c) *Histogram of Factor B*



(d) *Histogram of Factor C*



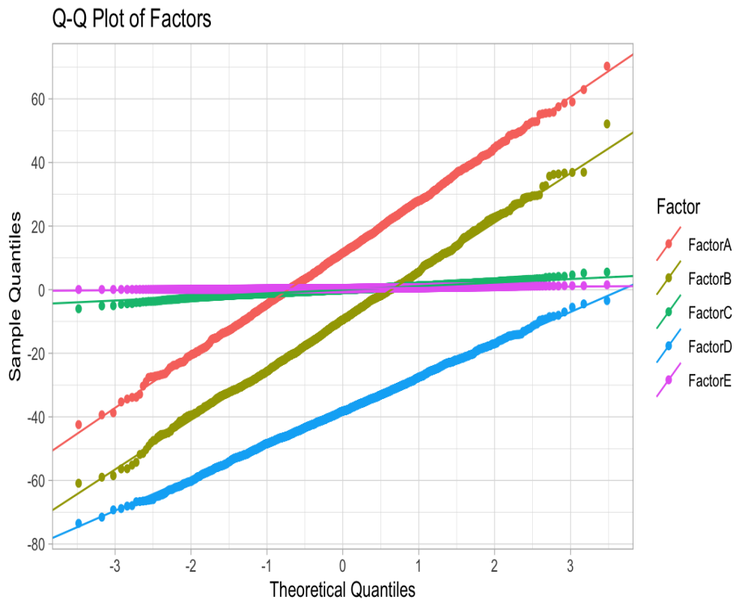
(e) *Histogram of Factor D*



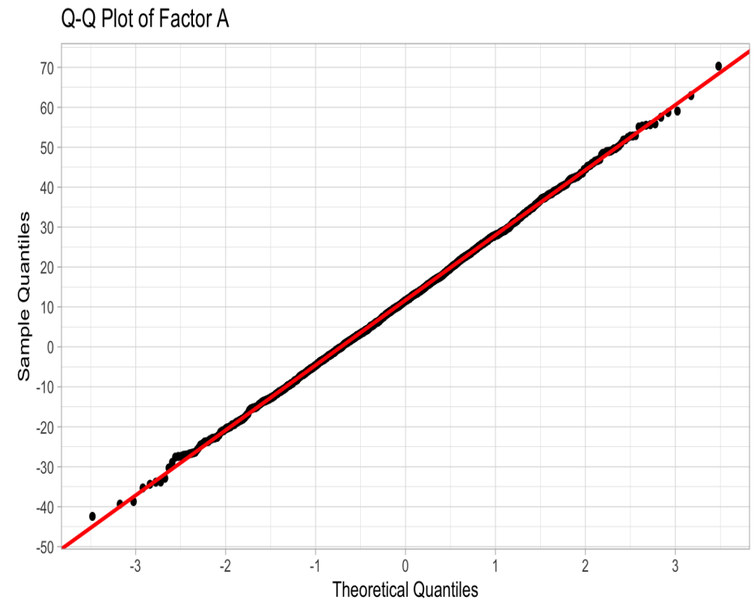
(f) *Histogram of Factor E*

Figure 2.1: *Histograms of Explanatory Variables*

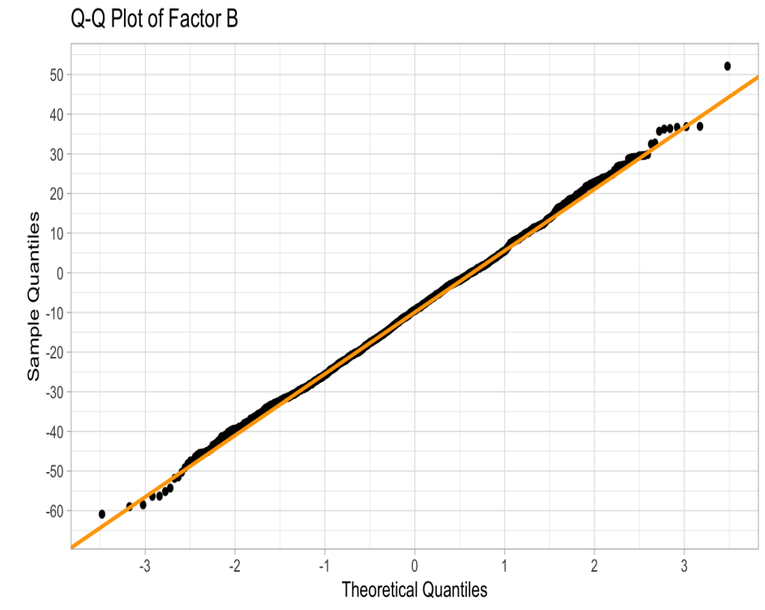
> ***Note***
>
> Bins are chosed based on Freedman–Diaconis (FD) rule, for robustness towards to outliners. Bins $\approx \frac{\max(x)-\min(x)}{h}$, where $h = 2\,\frac{IQR}{n^{1/3}}$, IQR = interquartile range of the data, n = sample size



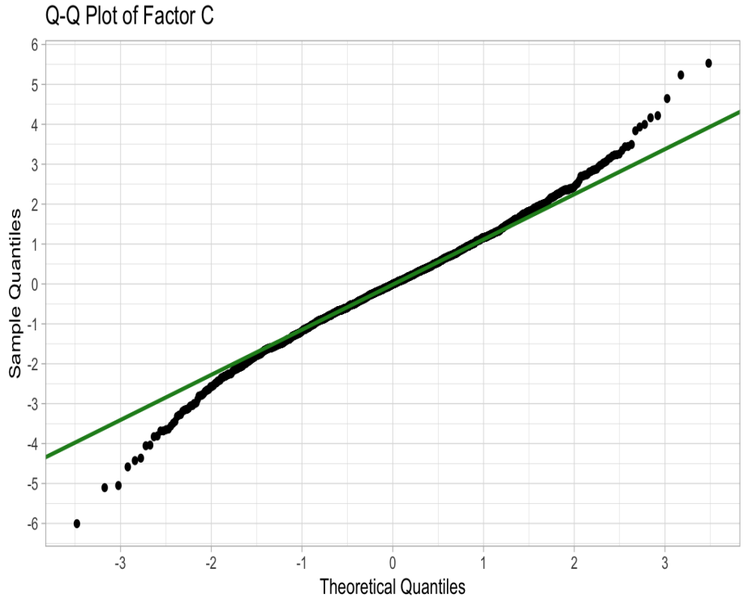
(a) *Quantitle-Quantile of Factors*



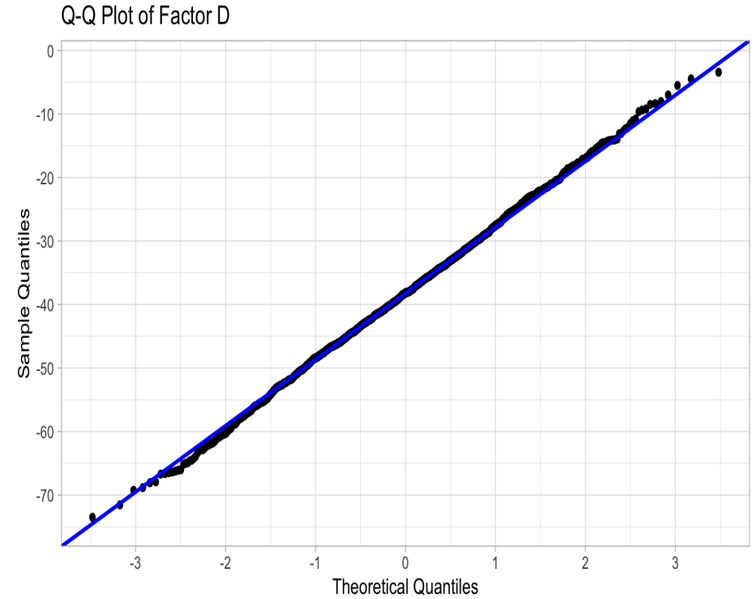
(b) *Quantitle-Quantitle plot of Factor A*



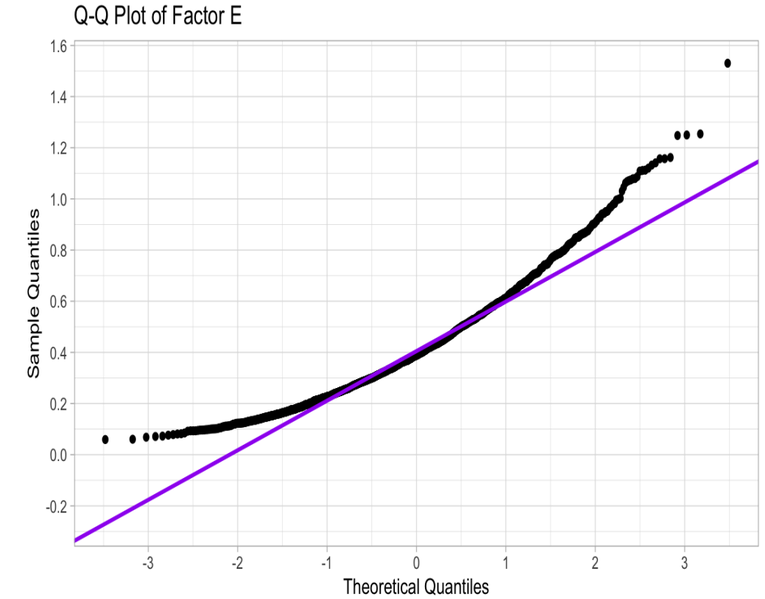
(c) *Quantitle-Quantitle plot of Factor B*



(d) *Quantitle-Quantitle plot of Factor C*



(e) *Quantitle-Quantitle plot of Factor D*



(f) *Quantitle-Quantitle plot of Factor E*

Figure 2.2: *Quantile Quantile plots of Explanatory Variables*

Inference

From the Quantitle-Quantitle plots, we can observe data points of Factor Afigure 2.2b, Factor figure 2.2c and Factor figure 2.2e are almost close to the quantile line. So, it is safe to assume that Factor A,B,D are normally distributed. Where as other factor has large deviations, so they are not normally distributed. We have confirmed the same from Shapiro-Wilk normality test table 2.1.

2.2 Simple Linear Regression Models

Now we consider each Factor in the explanatory variables and regress it against the Yvariable individually, so that we can understand the importance of that particular factor in determining the Yvariable outcome. The SLR model would be in the form, Yvariable $= a_0 + a_1$Factor $+ \epsilon_i$. We particularly look at the Adjusted R^2-squared value and p-value of each model. The accepting or rejecting the null-hypothesis (H_0):$a_1 = 0$ depending on the value of p. The summary for each SLR model is given as follows,

Factor A

	Dependent variable: Yvariable
Factor A	0.083 (0.125)
Constant	30.572*** (2.493)
Observations	2,000
R^2	0.0002
Adjusted R^2	−0.0003
F-statistic: 0.4403	p-value: 0.507
Note:	*p<0.1; **p<0.05; ***p<0.01

Factor B

	Dependent variable: Yvariable
Factor B	−4.602*** (0.079)
Constant	−12.560*** (1.450)
Observations	2,000
R^2	0.629
Adjusted R^2	0.629
F-statistic: 3390	p-value: $< 2.2 \times 10-16$
Note:	*p<0.1; **p<0.05; ***p<0.01

Table 2.2: *Factor A and Factor B Regression Results*

As we can see from table 2.2, Factor A has R^2 value to be $2 \times 10-4$ which is very insignificant value as R^2 is the measure of goodness of fit. The Adjusted R^2-squared value is found to be negative which indicates that using the Factor A is worse than using no explanatory variable at all. The p-value also suggests us to **accept the** H_0. So, we can conclude that **Factor A** is **not statistically significant** in determining the outcome of Yvariable.

From the regression results of Factor B, we can both R^2 and adjusted R^2-squared has the same value, which is 0.629. The Standard Deviation (Std Dev) error of factor B is also very small. From the p-value, we can understand that we have to **reject the** H_0. Thus we can conclude that **Factor B** is **statistically significant** in determining the outcome of Yvariable.

Factor C

	Dependent variable: Yvariable
Factor C	−4.524*** (1.624)
Constant	31.463*** (2.026)
Observations	2,000
R^2	0.004
Adjusted R^2	0.003
F-statistic: 7.757	p-value: 0.0054
Note:	*p<0.1; **p<0.05; ***p<0.01

Factor D

	Dependent variable: Yvariable
Factor D	4.419*** (0.163)
Constant	200.420*** (6.481)
Observations	2,000
R^2	0.268
Adjusted R^2	0.268
F-statistic: 731.7	p-value: $< 2.2 \times 10^{-16}$
Note:	*p<0.1; **p<0.05; ***p<0.01

Table 2.3: *Factor C and Factor D Regression Results*

From table 2.3 we can observe that, both the adjusted R^2-squared and R^2 values are found to be in the magnitude of 10^{-3}. But, from the value of p, we have to **reject the** H_0. So, we can conclude that statistically, **Factor C** is **statistically significant** in determining the outcome of Yvariable.

Both the adjusted R^2-squared and R^2 values of Factor D is found to be 0.268 and the Std Dev error is also minimal indicating that. Based on the p-value, we **reject the** H_0. So, the **factor D** alone is not completely explain the whole Yvariable but it is **statistically significant** in determining the outcome.

From the table 2.4, we can observe that both the adjusted R^2-squared and R^2 values of Factor E are found to be in the magnitude of 10^{-3} and also Std Dev error is by far the largest among the explanatory variables. We can also

accept the H_0 based on the p-value. So it is safe to conclude that **Factor E** is **not statistically significant** in determining the outcome of Yvariable.

	Dependent variable: Yvariable
Factor E	−15.191 (10.034)
Constant	37.938*** (4.692)
Observations	2,000
R^2	0.001
Adjusted R^2	0.001
F-statistic: 2.292	p-value: 0.1302
Note:	*p<0.1; **p<0.05; ***p<0.01

Table 2.4: *Regression Results on Factor E*

Thus we can conclude that among all the explanatory variables, only the **Factor B**, **Factor C** and **Factor D** are important and can explain the Yvariable significantly.

2.3 Multi Variable Linear Regression Model

The MLR model with all the explanatory variables is given by,

$$\text{Yvariable}_i = a_0 + a_1\text{FactorA}_i + a_2\text{FactorB}_i + a_3\text{FactorC}_i + a_4\text{FactorD}_i + a_5\text{FactorE}_i + \epsilon_i \quad (2.1)$$

$$Y = \beta\mathbb{X} + \epsilon$$

$$Y = \begin{pmatrix} \text{Yvariable}_1 \\ \vdots \\ \text{Yvariable}_n \end{pmatrix} \quad \mathbb{X} = \begin{pmatrix} 1 & \text{FactorA}_1 & \cdots & \text{FactorE}_1 \\ \vdots & \vdots & & \vdots \\ 1 & \text{FactorA}_n & \cdots & \text{FactorE}_n \end{pmatrix} \quad \boldsymbol{\beta} = \begin{pmatrix} a_0 \\ \vdots \\ a_5 \end{pmatrix} \quad \boldsymbol{\epsilon} = \begin{pmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{pmatrix} \quad (2.2)$$

Variance Inflation Factor (VIF): The VIF is a parameter used to detect multicollinearity among the explanatory variables in the MLR model, which is given by

$$\text{VIF} = \frac{1}{1 - R_j^2}$$

Where, R_j^2 is the Adjusted R^2-squared value for the linear regression on explanatory variables given by

$$X_j = \alpha + \beta_1 X_1 + \ldots + \beta_{j-1}X_{j-1} + \beta_{j+1}X_{j+1} + \ldots + \beta_p X_p + \varepsilon$$

VIF value Interpretation:

- ~ 1: No multicollinearity
- 1-5: Low to moderate collinearity and usually not a concern
- 5-10: Moderate to high collinearity and further analysis needed
- ~ 10: Serious multicollinearity and it is problematic

Factor A	Factor B	Factor C	Factor D	Factor E
1.001577	1.004196	1.001915	1.000617	1.002353

Table 2.5: *VIF values of Factors*

From the table 2.5, we can see that VIF values for all the factors are close to 1, so we can conclude that there is **no collinearity** between the explanatory variables and we can include all the factors in the MLR model.

From table 2.7, we can observe that both the adjusted R^2-squared and R^2 values of MLR model with all factors, are found to be 0.906 which indicates that the model is the best fit for Yvariable. Still the Std Dev of Factor E is considerably larger when compared with other factors Std Dev. Though the R^2 indicating that we have a really good fit for the model, we still haven't investigated into the multicollinearity of the explanatory variables. The BIC for this MLR model is found to be 19021.98

So, we now conduct our analysis into the Variance Inflation Factor for multicollinearity and BIC score for other models, as we need lower the BIC score indicated the better the model.

2.3.1 Best Model

To find the best model, there are two parameters that we can look at, which are R^2-squared value and BIC score.

Bayes Information Criterion: Bayes Information Criterion is a criterion for choosing the best model among several models that can be possible by adding or removing the explanatory variables. It is based on likelihood function to an extent and closely similar to Akaike Information Criterion (AIC).

We use the step function[6] available in the R programming language to find the best model based on BIC.We can see from the results table 2.6, the BIC scores for different models. The model with Factors B,C,D has the lowest BIC score. However the scores are relatively close to each other, we pick the model with lowest score.

Model	**BIC Score**
$\text{Yvariable}_i = a_0 + a_1\text{FactorA}_i + a_2\text{FactorB}_i + a_3\text{FactorC}_i + a_4\text{FactorD}_i + a_5\text{FactorE}_i + \epsilon_i$	13305.02
$\text{Yvariable}_i = a_0 + a_2\text{FactorB}_i + a_3\text{FactorC}_i + a_4\text{FactorD}_i + a_5\text{FactorE}_i + \epsilon_i$	13303.84
$\text{Yvariable}_i = a_0 + a_2\text{FactorB}_i + a_3\text{FactorC}_i + a_4\text{FactorD}_i + \epsilon_i$	**13302.91**

Table 2.6: *BIC Score of different models*

Table 2.7: *Multi Regression Results with all Factors*

	Dependent variable: Yvariable
FactorA	−0.035 (0.038)
FactorB	−4.629*** (0.040)
FactorC	−1.947*** (0.499)
FactorD	4.488*** (0.058)
FactorE	3.242 (3.077)
Constant	157.720*** (2.693)
Observations	2,000
R^2	0.906
Adjusted R^2	0.906
Residual Std. Error	27.792 (df = 1994)
F Statistic	3,865.646*** (df = 5; 1994)
Note:	*p < 0.1; **p < 0.05; ***p < 0.01

Table 2.8: *Multi Regression Results of Best Model*

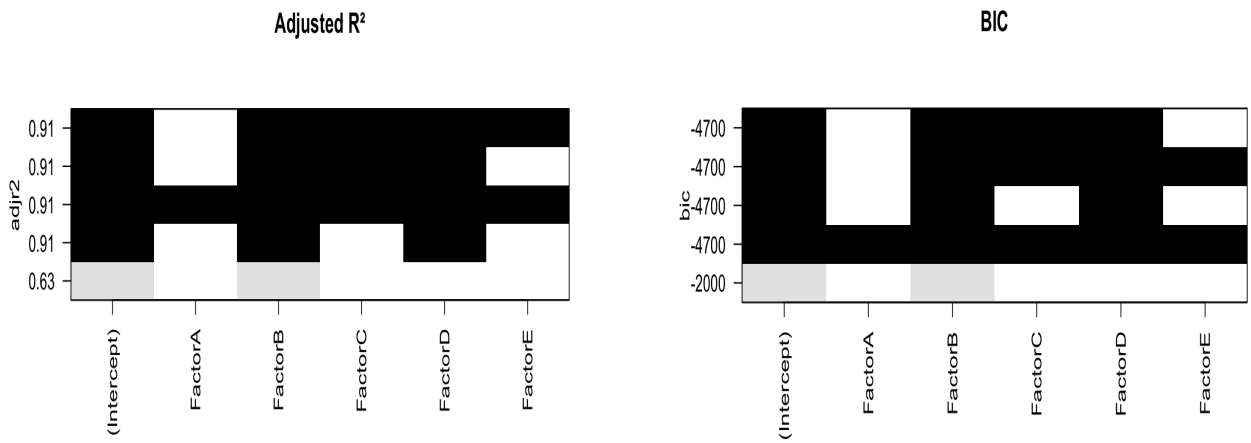
	Dependent variable: Yvariable
FactorB	−4.626*** (0.040)
FactorC	−1.940*** (0.499)
FactorD	4.488*** (0.058)
Constant	158.674*** (2.346)
Observations	2,000
R^2	0.906
Adjusted R^2	0.906
Residual Std. Error	27.791 (df = 1996)
F Statistic	6,442.477*** (df = 3; 1996)
Note:	*p < 0.1; **p < 0.05; ***p < 0.01

Note

The BIC score mentioned in the table 2.6 has the scaled BIC score, which is not the true BIC score. The step function compares models using a relative criterion, but not absolute BIC. It drops the constant parameter in the model when evaluating as constant is irrelevant and dropping them improve stability and the scaled BIC score only for the comparison between the different models. The true BIC score is obtained using *BIC(model)* command in R.

The true BIC score of the best fit model, $\text{Yvariable}_i = a_0 + a_2\text{FactorB}_i + a_3\text{FactorC}_i + a_4\text{FactorD}_i + \epsilon_i$ is found to be, BIC = 19008.67 which is lower than the BIC score of 19021.98 for the MLR model with all factors.

Finding the best model based on the adjusted R^2-squared values is not a best practice in our case, as the values for different models are very close to each other, as can be see from table 2.8, table 2.7. By for visualization, we can observe the following figure 2.3, where the intensity of the black color indicates that the explanatory variable is more used in different models. From figure 2.3b, we can see that the model with Factors B,C,D is the best model among all other models.



(a) *Adjusted R^2 vs Model Size* **(b)** *BIC vs Model Size*

Figure 2.3: *Comparison of subset selection metrics for linear regression models*

2.4 Prediction using MLRM

The MLR model with significant explanatory variables is given by,

$$\text{Yvariable} = 158.67411 - 4.62612\ \text{FactorB} - 1.94034\ \text{FactorC} + 4.48751\ \text{FactorD} \tag{2.3}$$

For the observation, Factors = (34,76,-25,-55,42), we get the Yvariables as,

$$\begin{aligned}\text{Yvariable} &= 158.67411 - 4.62612 \times 76 - 1.94034 \times -25 + 4.48751 \times -55 \\ &= 158.67411 - 351.58512 + 48.5085 - 246.81305 = -391.21556\end{aligned}$$

$$\boxed{\text{Yvariable} = -391.2159}$$

Therefore, the company profits (negative Yvaraible) from the given observations.

2.5 Conclusion

From our analysis, with the given data, we have concluded that Factors A,B,D are normally distributed and the Factors C,E does not follow a normal distributed. Based on our Simple linear regression(SLR) model for each factor, we have observed that Factor A, Factor E has least importance on the determining the outcome of Yvariable and Factors B,C,D are more statistically more insignificant.

The Multi linear regression (MLR) model with Factors B,C,D as the explanatory variables with BIC score = 19008.67 is a better model than the MLR model with all the factors as explanatory variables with BIC score = 19021.98. Lastly, we have predicted the Yvariable to be -391.2159 from the observation, Factors = (34,76,-25,-55,42). Further more we can extend our study, to the errors terms (ϵ), we can perform tests like Breusch-Pagen Godfrey (BPG) test, or White general Heteroskedasticity test to support our assumption that error terms has constant error variance and they are biased to any explanatory variables. Since, our data is not time-series data, we are not required to test for Autocorrelation.

Bibliography

1. Grigorova, M. A very short introduction to risk management (2024).
2. Hlavac, M. *stargazer: Well-Formatted Regression and Summary Statistics Tables* Vignette and reference manual available at: https://cran.r-project.org/web/packages/stargazer/vignettes/stargazer.pdf and https://cran.r-project.org/web/packages/stargazer/refman/stargazer.html. CRAN (2022).
3. McNeil, A. J., Frey, R. & Embrechts, P. *Quantitative Risk Management: Concepts, Techniques and Tools* Revised edition. 699 pp. ISBN: 978-0-691-16627-8 (Princeton University Press, Princeton Oxford, 2015).
4. Meucci, A. *Risk and Asset Allocation* (Springer, New York, 2005).
5. Plotly Technologies Inc. *Plotly for R* https://plotly.com/r/. Accessed: 18 Dec 2025. 2025.
6. R Project. *Step Function* https://search.r-project.org/CRAN/refmans/emdi/html/step.html. Accessed: 2025-12-20. n.d.

Theorems and Derivations

Slutsky' s theorem: Let Y_n be a sequence of random variables which converges to Y in distribution and (K_n) be a sequence of random variables which convergences in probability **to a constant** k. Then, the following convergences hold:

- $(K_n + Y_n)$ converges in distribution to $k + Y$.
- $(K_n Y_n)$ converges in distribution to kY.
- If $k \neq 0$, then (Y_n/K_n) converges in distribution to Y/k.

Application of Slutsky's Theorem to find the Confidence Interval of Parameter μ: From section 1.1.2.2 and section 1.1.2.1, we can write,

$$\frac{\sqrt{k\Delta}(\widehat{\mu} - \mu)}{\sigma} \sim \mathcal{N}(0,1) \qquad \text{and} \qquad \frac{\widehat{\sigma}}{\sigma} \xrightarrow[k \to +\infty]{\text{probability}} 1$$

$$\frac{\sqrt{k\Delta}(\widehat{\mu} - \mu)}{\sigma} \times \frac{\sigma}{\widehat{\sigma}} \sim \mathcal{N}(0,1) \qquad \text{Using Third property of Slutsky's Theorem} \tag{A.1}$$

$$\text{Thus we have,} \quad \frac{\sqrt{k\Delta}(\widehat{\mu} - \mu)}{\widehat{\sigma}} \sim \mathcal{N}(0,1)$$

χ^2-Distribution: Let $Z_1, Z_2 ..., Z_k$ be k independent identically distributed random variables following the $\mathcal{N}(0,1)$. Then, the sum of the squares:

$$Z_1^2 + Z_2^2 ... + Z_k^2$$

follows the χ^2-distribution with k degrees of freedom.

$$S^2 = \frac{1}{k-1} \sum_{i=1}^{k} (X_i - \bar{X})^2 \qquad (k-1)S^2 = \sum_{i=1}^{k} (X_i - \bar{X})^2$$

$$\text{If} \quad X_i \sim \mathcal{N}(\text{Mean}, \text{Variance}) \quad \text{then,} \quad \frac{\sum_{i=1}^{k} (X_i - \bar{X})^2}{\text{Variance}} \sim \chi_k^2$$

$$\text{Therefore,} \quad \frac{(k-1)S^2}{\text{Variance}} \sim \chi_k^2 \tag{A.2}$$

Sample Mean: We are given $X_1, X_2, \ldots, X_n$ i.i.d random variables with finite second moment. Let's denote by $m := E(X_1)$

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^{n} X_i \qquad \mathbb{E}[\bar{X}_n] = \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^{n} X_i\right] \qquad = \frac{1}{n} \mathbb{E}\left[\sum_{i=1}^{n} X_i\right] \qquad \text{(linearity of expectation)}$$

$$= \frac{1}{n} \sum_{i=1}^{n} \mathbb{E}(X_1) = \frac{1}{n} \cdot n\, \mathbb{E}[X_1] = m \qquad \text{(i.i.d. assumption)}$$

$$\Rightarrow \boxed{\mathbb{E}[\bar{X}_n] = \mathbb{E}[X_1]} \tag{A.3}$$

$$\mathrm{Var}(\bar{X}_n) = \mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) \quad = \left(\frac{1}{n}\right)^2 \mathrm{Var}\left(\sum_{i=1}^{n} X_i\right) \quad = \left(\frac{1}{n}\right)^2 \sum_{i=1}^{n} \mathrm{Var}(X_i) \qquad \text{(independence)}$$

$$= \left(\frac{1}{n}\right)^2 \sum_{i=1}^{n} \mathrm{Var}(X_1) \qquad \text{(identical distribution)} \qquad = \left(\frac{1}{n}\right)^2 \cdot n \cdot \mathrm{Var}(X_1) \qquad = \frac{\mathrm{Var}(X_1)}{n}$$

$$\Rightarrow \boxed{\mathrm{Var}(\bar{X}_n) = \frac{\mathrm{Var}(X_1)}{n}} \tag{A.4}$$

Sample Variance

$$V_n := \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X}_n)^2.$$

$$(X_i - \bar{X}_n)^2 = \left[(X_i - m) - (\bar{X}_n - m)\right]^2 = (X_i - m)^2 - 2(X_i - m)(\bar{X}_n - m) + (\bar{X}_n - m)^2.$$

Substituting into the definition of V_n, we get

$$V_n = \frac{1}{n}\sum_{i=1}^{n}(X_i - m)^2 - \frac{2}{n}(\bar{X}_n - m)\sum_{i=1}^{n}(X_i - m) + \frac{1}{n}\sum_{i=1}^{n}(\bar{X}_n - m)^2$$

$$= \frac{1}{n}\sum_{i=1}^{n}(X_i - m)^2 - 2(\bar{X}_n - m)\,(\bar{X}_n - m) + (\bar{X}_n - m)^2,$$

Therefore,

$$\boxed{V_n = \frac{1}{n}\sum_{i=1}^{n}(X_i - m)^2 - (\bar{X}_n - m)^2}$$

We compute the expectation of V_n.

$$\mathbb{E}[V_n] = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n}(X_i - m)^2\right] - \mathbb{E}\left[(\bar{X}_n - m)^2\right] = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\left[(X_i - m)^2\right] - \mathbb{E}\left[(\bar{X}_n - m)^2\right] \qquad \text{(linearity of expectation)}$$

$$\mathbb{E}[V_n] = \frac{1}{n}\sum_{i=1}^{n}\mathrm{Var}(X_i) - \mathrm{Var}(\bar{X}_n), \qquad \mathrm{Var}(X_i) = \sigma^2, \qquad \mathrm{Var}(\bar{X}_n) = \frac{\sigma^2}{n}$$

Therefore,

$$\boxed{\mathbb{E}[V_n] = \sigma^2 - \frac{\sigma^2}{n} = \left(\frac{n-1}{n}\right)\sigma^2} \tag{A.5}$$

The estimator V_n is biased for $\mathrm{Var}(X_i)$. If V_n were unbiased, we would have $\mathbb{E}[V_n] = \mathrm{Var}(X_i)$ which is not the case. So, let's define

$$S_n^2 := \frac{n}{n-1}V_n$$

Then

$$\mathbb{E}\left(S_n^2\right) = \mathbb{E}\left(\frac{n}{n-1}V_n\right) \qquad = \frac{n}{n-1}\mathbb{E}(V_n)$$

$$= \frac{n}{n-1}\cdot\frac{n-1}{n}\mathrm{Var}(X_i) \qquad = \mathrm{Var}(X_i).$$

Hence, $S_n^2 = \frac{n}{n-1}\nu_n$ is an unbiased estimator of $\mathrm{Var}(X_i)$. S_n^2 is called Adjusted Sample Variance, which is given by

$$S_n^2 = \frac{n}{n-1}\nu_n = \frac{n}{n-1}\cdot\frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X}_n)^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X}_n)^2 \tag{A.6}$$

Confidence Intervals Let Z_n be a random variable following $\mathcal{N}(0,1)$ distribution, the confidence interval for Z_n can be given as

$$\mathbf{P}(Z_n \leqslant y) = F_Z(y), \qquad \text{and} \qquad \mathbf{P}(Z_n \leqslant x) = F_Z(x),$$

$$\mathbf{P}(x < Z_n \leqslant y) = F_Z(y) - F_Z(x) \quad \Rightarrow \quad \mathbb{P}\left[q_z\left(\frac{\alpha}{2}\right) \leqslant Z_n \leqslant q_z\left(1-\frac{\alpha}{2}\right)\right] = 1-\alpha$$

Where,

$$F_Z(y) = 1 - \frac{\alpha}{2} \quad , \quad F_Z(x) = \frac{\alpha}{2} \quad \text{and} \quad x = F_Z^{-1}\left(\frac{\alpha}{2}\right) \quad , \quad y = F_Z^{-1}\left(1-\frac{\alpha}{2}\right)$$

Let q_z is the inverse function of F_z (Cumulative Distribution Function (CDF)), called as quantile function of distribution of Z_n. The Quantitle function for standard normal distribution is symmetric in nature about 0, so we get

$$q_z\left(1-\frac{\alpha}{2}\right) = -q_z\left(\frac{\alpha}{2}\right)$$

Therefore, the confidence interval of Z_n is given by,

$$\boxed{\mathbf{P}\left[-q_z\left(1-\frac{\alpha}{2}\right) \leqslant Z_n \leqslant q_z\left(1-\frac{\alpha}{2}\right)\right] = 1-\alpha} \tag{A.7}$$

Loss Operator for the simulation of Value at Risk: From the equation (1.1), we get

$$X_k = \ln\left(\frac{S_t}{S_{t-\Delta}}\right) \quad \Rightarrow \quad S_t = e^{X_t}S_{t-\Delta}$$

Since we are using only one asset and one weight, S_t is same as V_t, the loss of the wealth is given by,

$$\begin{aligned} L_t &= -(V_{t+\Delta} - V_t) &&= -(V_t \cdot e^{X_k} - V_t) \\ &= -V_t\left(e^{X_k} - 1\right) &&= -S_t(1 - e^{X_k}) \end{aligned}$$

$$L_t = -S_t(e^{X_k} - 1) \qquad \text{and Loss operator:} \qquad e^{X_k} - 1 \tag{A.8}$$

B Problem 1:R Script

```
### ---------------- Clear all Outputs and Terminal of R Studio ------------------
## This for safety and to avoid variables conflict
rm(list = ls())
graphics.off()
cat("\014")

### --------------------- Installing Packages Required --------------------------
options (repos = "https://cran.rstudio.com")
packages <- c ( "ggplot2","readxl","plotly")
installed_packages <- packages %in% rownames(installed.packages())

# Install packages,if  not yet installed
if (any(installed_packages == FALSE)) {
  install.packages ( packages [!installed_packages])
}
invisible (lapply(packages,library,character.only = TRUE))   # Package Loading


### ----------------- 1. Stocks prices and Log returns on Stock ------------------
### ------------------------------------------------------------------------------

### Loading Data
getwd()
prices <- read_excel("Problem1set2.xlsx")

head(prices)
print(prices[,1])
print(prices[,2])         # Print out the vector of  "prices" of stock A

### Nnumber of Columns
ncol(prices)
###number of rows (that is our n in the lectures)
n_obs = nrow(prices)

##### We use the function summary()
summary(prices$"A")
mean(prices$"A")
#### The adjusted empirical variance for stock A
var(prices$"A")

###------------------------ Visualization using ggplot ---------------------------
p <- ggplot(prices, aes(x=Date, y=A)) +
  geom_area(fill="#69b3a2", alpha=0.4) +
  geom_line(color="#69b3a2") +
```

```
  labs(x = "Consecutive Days",y = "Stock Prices",title = "Stock A daily prices")
p   ## Displaying Plot

### ---------------------------- Returns on Stock -------------------------------
#We create a numeric vector of size n-1
returnsA<-vector(mode = "numeric", length = n_obs -1)
#then, we use a "for loop" in R
for(i in 1:n_obs-1) returnsA[i]<-(prices$`A`[i+1]-prices$`A`[i])/prices$`A`[i]
# Plot for returns
plot(returnsA, type='l', xlab="Days", ylab="Returns on Stock")

### ------------------------ Logarithm Returns on Stock A ------------------------
log_returns <- diff(log(prices$`A`), lag=1)

# Plot for Log returns
df <- data.frame(Date = 1:length(log_returns),LogReturn = log_returns)
q <- ggplot(df, aes(x=Date, y=log_returns)) +
  geom_line(color="steelblue") +
  labs(x = "Days", y = "Log returns on Stock A",title = "Log Returns")
q

### ---------------------------- Histogram -------------------------------------
hist(log_returns, probability = TRUE, breaks = "FD",
     main = "Log Returns", xlab = "Value", ylab = "Density")
lines(density(log_returns), lwd = 2)

x <- seq(min(log_returns), max(log_returns), length.out = 200)
lines(x, dnorm(x, mean(log_returns), sd(log_returns)), col = "forestgreen",lwd = 2)

legend("topright", legend = c("Kernel Density Estimate", "Normal Fit"),
       lwd = c(2, 2), col = c("black", "forestgreen"), bty = "n")

# Split plotting area, 1 row 2 columns
par(mfrow = c(1, 2), mar = c(4, 4, 2, 1))

first_half <- log_returns[1:floor(n_obs/2)]
second_half <- log_returns[(floor(n_obs/2)):n_obs-1]

# First Half
hist(first_half,probability = TRUE,breaks = 20,
     main = "Log Returns (First Half)", xlab = "Value", ylab = "Density",
     ylim = c(0, max(density(first_half)$y, dnorm(first_half, mean(first_half),
     sd(first_half)))))

lines(density(first_half), lwd = 2, col = "black")
xgrid1 <- seq(min(first_half), max(first_half), length.out = 200)
lines(xgrid1, dnorm(xgrid1, mean(first_half), sd(first_half)), col = "red", lwd = 2)

# Second half
hist(second_half,probability = TRUE,breaks = 20,
     main = "Log Returns (Second Half)",xlab = "Value",ylab = "Density",
     ylim = c(0, max(density(second_half)$y, dnorm(second_half, mean(second_half),
     sd(second_half)))))
```

```
lines(density(second_half), lwd = 2, col = "black")
xgrid2 <- seq(min(second_half), max(second_half), length.out = 200)
lines(xgrid2, dnorm(xgrid2, mean(second_half), sd(second_half)), col = "red", lwd = 2)
legend("topright",legend = c("Kernel Density", "Normal Fit"),col = c("black", "red"),
       lwd = 2,bty = "n")

par(mfrow = c(1,1)) ## Restoring the plotting to normal

### ------------------ Kolmogorov-Sirmnov Test for Same distribution -------------

## To check if both the data sets are coming from same distribution
ks.test(first_half, second_half)

### -------------------------- Quantile-Quantile Plots Plot -----------------------

qqplot <- ggplot(data.frame(log_returns), aes(sample = log_returns)) +
  stat_qq(shape = "*", size = 6) +
  stat_qq_line(color = "red", linewidth = 0.75) +
  labs(x = "Theoretical Quantiles",y = "Sample Quantiles",
       title = "Q-Q Plot of Log Returns") +
  scale_x_continuous(n.breaks = 10) +
  scale_y_continuous(n.breaks = 10) +
  theme_light()
qqplot

### --------------------------------- Meucci Visual Check ------------------------
### The shatter plot should be circular, if the data is i.i.d random variables

lA1<-vector(mode = "numeric", length = n_obs-1)
for(i in 1:n_obs-1) lA1[i]<-log_returns[i]

lA2<-vector(mode = "numeric", length = n_obs-1)
for(i in 1:n_obs-1) lA2[i]<-log_returns[i+1]

plot(lA1, lA2, xlab="Log Returns", ylab="Log Returns Lagged", asp=1)

### --------------------------- Mean, Variance and Covariance --------------------

X_mean  <- mean(log_returns) ## Log returns mean
X_adjvar <- var(log_returns) ## Log returns Adjusted Variance
cov_mat <- cov(as.matrix(log_returns))
cov_mat_GBM <- diag(X_adjvar, length(log_returns))
X_mean
X_adjvar
cov_mat

### -------------- Shapiro-Wilk Normality, Box-Pierce and Box-Ljung Tests -------
## Normality test
shapiro.test(log_returns)

## Box-Pierce test on the log-returns
Box.test (log_returns, lag = 5, type = "Box-Pierce")
```

```r
## Box-Ljung test on the log-returns
Box.test (log_returns, lag = 5, type = "Ljung-Box")

### ------------------------- Kolmogorov-Smirnov Test ----------------------------
## For Normal Distribution
ks.test(log_returns,pnorm,mean=mean(X_mean),sd=sd(log_returns))


### ----------------------------- Parameters -------------------------------------
Delta <- 1/252 # One trading day
mu_hat <- X_mean/Delta + X_adjvar/2
sigma_hat <- sqrt (X_adjvar/Delta)

mu_hat ## parameter mu
sigma_hat ## parameter Sigma


### ------------------------ Confidence Intervals ------------------------------
### Confidence Interval for mu
alpha <- 0.05
k = length(log_returns) # total log returns data points
se_mu <- sigma_hat / sqrt(k * Delta)
mu_hat + c(-1,1) * qnorm(1 - alpha/2) * se_mu

### Confidence Interval for sigma
lower <- (k-1) * sigma_hat^2 / qchisq(1 - alpha/2, df = k)
upper <- (k-1) * sigma_hat^2 / qchisq(alpha/2, df = k)
sqrt(c(lower, upper))  # Confidence Interval for sigma


### -------------------- 2. Losses on the stock and Value at Risk ---------------
### ----------------------------------------------------------------------------

### ------------------ Historical Method: Value at Risk (vaR) -------------------
St <- as.numeric(tail(prices$'A', 1))

loss_op <- -St * (exp(log_returns) - 1) ## Loss operator
losses_sorted <- sort(loss_op)

alpha <- 0.95
k <- length(loss_op)
VaR_hist <- losses_sorted[floor(k * alpha)]
VaR_hist
VaR_hist/St ## VaR in percentage


### -------------------------- Monte Carlo Simulation ----------------------------

alpha <- 0.999 # confidence level
N <- 100000
set.seed(24)   # for reproducibility
St <- as.numeric(tail(prices$'A', 1)) # Current portfolio value
```

```r
# Parameters
mu_hat <- mean(log_returns) / Delta + 0.5 * var(log_returns) / Delta
sigma_hat <- sd(log_returns) / sqrt(Delta)
sim_returns <- (mu_hat - 0.5 * sigma_hat^2) * Delta +
  sigma_hat * sqrt(Delta) * rnorm(N)

sim_losses <- - St * (exp(sim_returns) - 1)
sim_losses_sorted <- sort(sim_losses)
VaR_MC <- sim_losses_sorted[floor(N * alpha)]
VaR_MC
VaR_MC/St

### NOTE: change alpha values to 0.99 and 0.999 to get VaR simulation results for 99%
### and 99.9% confidence levels respectively
```

C Problem 2:R Script

```
### -------------------- Clear all Outputs and Terminal of R Studio ----------------
## This for safety and to avoid variables conflict
rm(list = ls())
graphics.off()
cat("\014")

### ---------------------- Installing Packages Required --------------------------
options (repos = "https://cran.rstudio.com")
packages <- c ("ggplot2","readxl","plotly","stargazer", "regclass","leaps", "car")
installed_packages <- packages %in% rownames(installed.packages())

# Install packages, if not yet installed
if (any(installed_packages == FALSE)) {
  install.packages ( packages [!installed_packages])
}
invisible (lapply(packages,library,character.only = TRUE))   # Package Loading

### ----------------------------- Loading Data ----------------------------------
## To get current directory
getwd()
obs <- read_excel("Problem2set2.xlsx")

### Number of Columns
ncol(obs)
### Number of rows
n_obs=nrow(obs)

### Assigning Variables to Data
head(obs)
Y_obs <- obs[, 1]$'Yvariable'
factors <- obs[, 2:6]

FactorA <- factors[, 1]$'FactorA'
FactorB <- factors[, 2]$'FactorB'
FactorC <- factors[, 3]$'FactorC'
FactorD <- factors[, 4]$'FactorD'
FactorE <- factors[, 5]$'FactorE'

### ------------------------- Shapiro-Wilk Normality Test ------------------------
tests <- list(
  "Factor A" = shapiro.test(FactorA),
  "Factor B" = shapiro.test(FactorB),
  "Factor C" = shapiro.test(FactorC),
  "Factor D" = shapiro.test(FactorD),
  "Factor E" = shapiro.test(FactorE)
```

```
)
tests

### ---------------------------- Histogram Plots --------------------------------

## Bins are choosed based on Freedman-Diaconis (FD) rule, for robustness towards to outliners

fig <- plot_ly(alpha = 0.6)
fig <- fig %>% add_histogram(x = FactorA, name = 'Factor A')
fig <- fig %>% add_histogram(x = FactorB, name = 'Factor B')
fig <- fig %>% add_histogram(x = FactorC, name = 'Factor C')
fig <- fig %>% add_histogram(x = FactorD, name = 'Factor D')
fig <- fig %>% add_histogram(x = FactorE, name = 'Factor E')
fig <- fig %>% layout(barmode = "overlay")
fig

### ------------------------------ Factor A Histogram ----------------------------
hist(FactorA,probability = TRUE,breaks = "FD",main = "Factor A",xlab = "Value")
lines(density(FactorA), lwd = 2)

x <- seq(min(FactorA), max(FactorA), length.out = 200)
lines(x,dnorm(x, mean(FactorA), sd(FactorA)),col = "red",lwd = 2)

legend("topright",legend = c("Kernel Density Estimate", "Normal Fit"),
       lwd = c(2, 2), col = c("black", "red"), bty = "n")

### ------------------------------ Factor B Histogram ----------------------------
hist(FactorB,probability = TRUE,breaks = "FD",main = "Factor B",xlab = "Value")
lines(density(FactorB), lwd = 2)

x <- seq(min(FactorB), max(FactorB), length.out = 200)
lines(x,dnorm(x, mean(FactorB), sd(FactorB)),col = "orange",lwd = 2)

legend("topright",legend = c("Kernel Density Estimate", "Normal Fit"),
       lwd = c(2, 2),col = c("black", "orange"),bty = "n")

### ------------------------------ Factor C Histogram ----------------------------
hist(FactorC,probability = TRUE,breaks = "FD",main = "Factor C",xlab = "Value")
lines(density(FactorC), lwd = 2)

x <- seq(min(FactorC), max(FactorC), length.out = 200)
lines(x,dnorm(x, mean(FactorC), sd(FactorC)),col = "forestgreen",lwd = 2)

legend("topright",legend = c("Kernel Density Estimate", "Normal Fit"),lwd = c(2, 2),
       col = c("black", "forestgreen"), bty = "n")

### ------------------------------ Factor D Histogram ----------------------------
hist(FactorD,probability = TRUE,breaks = "FD",main = "Factor D",xlab = "Value")
lines(density(FactorD), lwd = 2)

x <- seq(min(FactorD), max(FactorD), length.out = 200)
lines(x,dnorm(x, mean(FactorD), sd(FactorD)),col = "blue",lwd = 2)
```

```
legend("topright",legend = c("Kernel Density Estimate", "Normal Fit"),lwd = c(2, 2),
       col = c("black", "blue"),bty = "n")

### ----------------------------- Factor E Histogram ----------------------------
hist(FactorE,probability = TRUE,breaks = "FD",main = "Factor E",xlab = "Value")
lines(density(FactorE), lwd = 2)

x <- seq(min(FactorE), max(FactorE), length.out = 200)
lines(x,dnorm(x, mean(FactorE), sd(FactorE)),
      col = "purple",lwd = 2)

legend("topright",legend = c("Kernel Density Estimate", "Normal Fit"),
       lwd = c(2, 2),col = c("black", "purple"),bty = "n")

###---------------------------- Quantile-Quantile Plots --------------------------
qq_plot <- ggplot() +
  stat_qq(aes(sample = FactorA, color = "FactorA")) +
  stat_qq(aes(sample = FactorB, color = "FactorB")) +
  stat_qq(aes(sample = FactorC, color = "FactorC")) +
  stat_qq(aes(sample = FactorD, color = "FactorD")) +
  stat_qq(aes(sample = FactorE, color = "FactorE")) +
  stat_qq_line(aes(sample = FactorA, color = "FactorA")) +
  stat_qq_line(aes(sample = FactorB, color = "FactorB")) +
  stat_qq_line(aes(sample = FactorC, color = "FactorC")) +
  stat_qq_line(aes(sample = FactorD, color = "FactorD")) +
  stat_qq_line(aes(sample = FactorE, color = "FactorE")) +
  labs(
    x = "Theoretical Quantiles",
    y = "Sample Quantiles",
    color = "Factor",
    title = "Q-Q Plot of Factors"
  ) +
  scale_x_continuous(n.breaks = 10) +
  scale_y_continuous(n.breaks = 10) +
  theme_light()
qq_plot

### ----------------------------- QQ Plot Factor A ------------------------------
qqplotA <- ggplot(data.frame(FactorA), aes(sample = FactorA)) + stat_qq() +
  stat_qq_line(color = "red", linewidth = 1) +
  labs(x = "Theoretical Quantiles",y = "Sample Quantiles",title = "Factor A") +
  scale_x_continuous(n.breaks = 10) +
  scale_y_continuous(n.breaks = 10) +
  theme_light()
qqplotA

### ----------------------------- QQ Plot Factor B ------------------------------
qqplotB <- ggplot(data.frame(FactorB), aes(sample = FactorB)) +stat_qq() +
  stat_qq_line(color = "orange", linewidth = 1) +
  labs(x = "Theoretical Quantiles",y = "Sample Quantiles",title = "Factor B") +
  scale_x_continuous(n.breaks = 10) +
  scale_y_continuous(n.breaks = 10) +
  theme_light()
```

```
qqplotB

### ----------------------------- QQ Plot Factor C -------------------------------
qqplotC <- ggplot(data.frame(FactorC), aes(sample = FactorC)) + stat_qq() +
  stat_qq_line(color = "forestgreen", linewidth = 1) +
  labs(x = "Theoretical Quantiles",y = "Sample Quantiles",title = "Factor C") +
  scale_x_continuous(n.breaks = 10) +
  scale_y_continuous(n.breaks = 10) +
  theme_light()
qqplotC

### ----------------------------- QQ Plot Factor D -------------------------------
qqplotD <- ggplot(data.frame(FactorD), aes(sample = FactorD)) + stat_qq() +
  stat_qq_line(color = "blue", linewidth = 1) +
  labs(x = "Theoretical Quantiles",y = "Sample Quantiles",title = "Factor D") +
  scale_x_continuous(n.breaks = 10) +
  scale_y_continuous(n.breaks = 10) +
  theme_light()
qqplotD

### ----------------------------- QQ Plot Factor E -------------------------------
qqplotE <- ggplot(data.frame(FactorE), aes(sample = FactorE)) +
  stat_qq() +
  stat_qq_line(color = "purple", linewidth = 1) +
  labs(x = "Theoretical Quantiles",y = "Sample Quantiles",title = "Factor E") +
  scale_x_continuous(n.breaks = 10) +
  scale_y_continuous(n.breaks = 10) +
  theme_light()
qqplotE


### --------------------------- Simple Linear Regression Models ------------------

# Simple Linear Regression For Factor A
regA <- lm(Y_obs~FactorA)
summary(regA)

# Simple Linear Regression For Factor B
regB <- lm(Y_obs~FactorB)
summary(regB)

# Simple Linear Regression For Factor C
regC <- lm(Y_obs~FactorC)
summary(regC)

# Simple Linear Regression For Factor D
regD <- lm(Y_obs~FactorD)
summary(regD)

# Simple Linear Regression For Factor E
regE <- lm(Y_obs~FactorE)
summary(regE)
```

```
### ----------------------- Summary to LaTeX table -----------------------------

### Using stargazer package, we are getting the simple regression model summary as LaTeX table in the outp
### **Un-comment** and Change the parameter regA for different factors.
### Regression summary to LaTeX table

# stargazer(
#   regE,
#   type = "latex",
#   title = "Regression Results on Factor E",
#   label = "tab:regE_summ",
#   single.row = TRUE,
#   dep.var.labels = "Yvariable",
#   covariate.labels = c("Factor E")
# )

### ------------------------ Multi-Linear Regression Model ------------------------

multireg <- lm(Y_obs~FactorA+FactorB+FactorC+FactorD+FactorE)
summary(multireg)
anova(multireg)

# stargazer(multireg, type = "latex",
#           title = "Multi Regression Results",
#           label = "tab:multireg_summ",
#           single.row = TRUE)

### ------------------------- Multi-Collinearity Test on Factors -----------------

vif(multireg)

### ------------------------- Bayes Information Criterion ------------------------

BIC(multireg)

#### Choosing the best model using, step function, a stepwise model selection f
#### or regression models. k=2 for Akaike Information Criterion.
### direction= "both" meaning allowing both addition and removal of factors based on
### the score of model

best_model <- step(multireg, direction="both", k=log(n_obs))
summary(best_model)
BIC(best_model)

# stargazer(best_model, type = "latex",
#           title = "Multi Regression Results of Best Model",
#           label = "tab:bestmodel_summ",
#           single.row = TRUE)


### ------------------- Best Model based on Adjusted R\^2- Squared ----------------
```

```
regfit <- regsubsets(
  Y_obs ~ FactorA + FactorB + FactorC + FactorD + FactorE, data=obs,nvmax = n_obs)
summary(regfit)


adjr2_plot <- plot(regfit, scale="adjr2", col=gray(seq(0, 0.9, length = 10)))
title(main="Adjusted R²", font.main=2)
BIC_plot <- plot(regfit, scale="bic")
title(main="BIC", font.main=2)


### ----------------- Prediction using the Multi-Linear Regression Model ----------

# Creating a data frame of new factors and calling the column "new_factors".
###This naming of the column is important whenn using predict afterwards.
new_factors <- data.frame(
  FactorA = 34,
  FactorB = 76,
  FactorC = -25,
  FactorD = -55,
  FactorE = 42
)

# Predicting the new value of the Yvaribale
new_Y_obs <-predict(best_model,newdata=new_factors)
new_Y_obs
```

WARWICK
THE UNIVERSITY OF WARWICK